VANSH LAKHWANI

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PROFILE SUMMARY

AI/ML Engineer with hands-on experience building production-grade applications using modern AI tools, full-stack development, and end-to-end ML pipelines. Currently pursuing B.Tech in Artificial Intelligence with *a 9.33 CGPA*. Demonstrated expertise in developing intelligent systems through internships at *TATA Technologies* and *Uluka Systems*, focusing on effort estimation, road condition assessment, and mission-critical system validation. Passionate about human-AI interaction and creating AI-driven solutions that optimize workflows and enhance user experiences.

TECHNICAL SKILLS

Languages: Python, C, R, TypeScript

AI/ML & Data Science: TensorFlow, Keras, NumPy, Pandas, Scikit-learn, OpenCV, LSTM, NLP

Frontend: React, Next.js 14/16, HTML/CSS, Tailwind CSS, Framer Motion

Backend & APIs: Django, Node.js, AuthO, Supabase, Groq AI, OpenAI API

Databases: PostgreSQL, MongoDB, MySQL, Redis

Tools & Platforms: Git/GitHub, Vercel, Firebase, Expo, LAMP Stack, Cal.com, ChatGPT, Claude, Ollama, Antigravity, Stitch

PROFESSIONAL EXPERIENCE

Project Intern, TATA Technologies, Pune **November 2025 – Present**

- Developing AI-driven Effort Estimator system to optimize software development timeline planning for enterprise stakeholders.
- Leveraging *transformer-based* models with embeddings to interpret unstructured requirements and generate structured *SDLC outputs*.
- Integrating *COCOMO model* with historical production data and resource allocation patterns to improve estimation reliability by 30%.

Software Engineer Intern; Uluka System (OPS) Pvt Ltd., Pune **October 2025**

- Validated mission-critical ADS-B electronic systems through comprehensive testing using *C, Python, and LAMP stack*.
- Conducted Factory Acceptance Testing (FAT) including BOM verification, functional validation, and GPS/NTP synchronization testing
- Performed *EMI/EMC* testing and resolved interference issues through advanced grounding and shielding techniques

Machine Learning Intern; Uluka System (OPS) Pvt Ltd., Pune **August 2024 – February 2025**

- Architected end-to-end ML pipeline for Road Condition Assessment using real-time sensor data from Android devices, processing 10,000+ data points
- Reduced feature dimensionality by 40% using PCA while engineering meaningful features (total acceleration, jerk, speed) from raw time-series signals
- Implemented K-Means clustering achieving 85% accuracy in classifying road surfaces into smooth, rough, and mixed categories with geospatial visualization via Folium

KEY PROJECTS

Prio - Task Management Platform **April 2026**

- Built production-grade real-time task manager using *Next.js 16, React 19, and Supabase* (*PostgreSQL, Realtime, Auth) with Google Calendar* integration
- Implemented interactive Kanban boards with *dnd-kit* drag-and-drop, *Zustand* state management, and *Zod* validation
- Designed responsive UI with *Tailwind CSS and Framer Motion*, featuring analytics dashboards via Recharts and rich text editing using *Tiptap.*

AI-Powered Portfolio Website **March 2026**

- Engineered full-stack portfolio with *Next.js 14, React, and TypeScript* featuring serverless API routes and AI chatbot integration (*Groq AI/OpenAI AP*I)
- Implemented end-to-end scheduling workflow using *Cal.com API* for automated Google Meet booking and GitHub API for dynamic project updates
- Integrated *Nodemailer* for contact functionality and designed responsive UI with *Tailwind CSS and Framer Motion*, deployed on *Vercel*

Smart CCTV Surveillance System **March – May 2024**

- Led team of 4 to engineer ML-powered surveillance system using Python and OpenCV for real-time CCTV footage analysis
- Enabled real-time crowd monitoring, face recognition, and anomaly detection achieving 90% accuracy in threat identification

Comment Toxicity Detection Model **February – April 2024**

- Developed deep learning *NLP model* using *TensorFlow* and *Keras* with *bidirectional LSTM* architecture for multi-label classification
- Trained on *Jigsaw* dataset to categorize comments into toxic, obscene, insult, and threat categories with 87% accuracy, deployed on *Gradio* interface

EDUCATION

Bachelor of Technology in Artificial Intelligence and Data Science

Dr. D.Y. Patil School of Science and Technology (DPU), Pune

CGPA: 9.33/10.0 **September 2022 - June2026**(*expected*)

CERTIFICATIONS & TRAINING

- Microsoft Azure Fundamentals **February 2025**
- Wadhwani Foundation - Entrepreneurship Program **September – December 2024**
- Skill4Future - Artificial Intelligence **October 2024**
- TCS Ion Career Edge - Young Professional **January 2024**
- AWS Conference **June 2023**

ACHIEVEMENTS & LEADERSHIP

- 3rd Place, 5G Nexus Ideathon – DPU, Pune **April 2024**
- Team Lead, Smart India Hackathon – DPU, Pune **September 2023**
  - Led team to build AI-driven CCTV system with real-time face recognition
- Vice President, Binary Brain Coding Club – DPU, Pune **December 2022 – October 2023**
  - Led coding workshops and mentored 50+ students